



UNIVERSITY OF AMSTERDAM
Amsterdam Business School

Data Science in Online Marketing

Course Revision

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Data Science in Online Marketing Course

Get a deep dive on concepts and algorithms used in online marketing:

Online Advertising

- CTR prediction
- Media Mix Modeling & Conversion Attribution



Recommenders

- Content Based
- Collaborative Filtering



amazon

NETFLIX



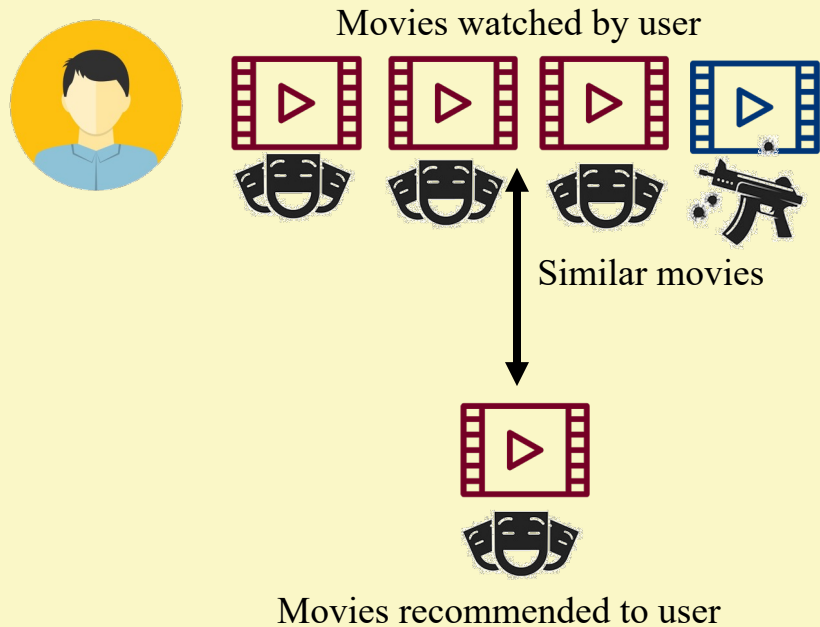


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Recommenders

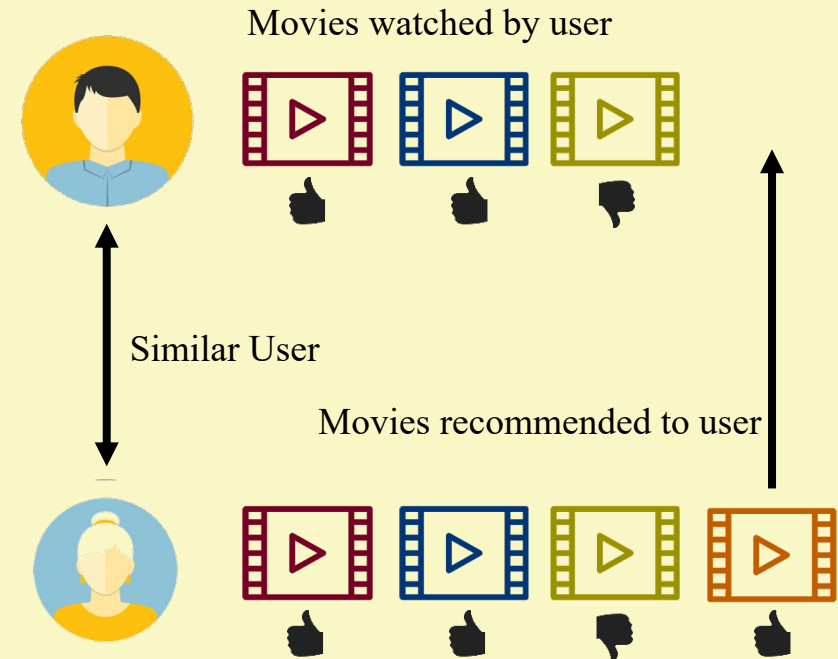
Content-based vs Collaborative Filtering

Content-based



Similarity based on **items characteristics**: e.g., genre

Collaborative Filtering



Similarity based on **users taste** (feedback)

User feedback

- Main data source: Rating/Feedback Matrix

- ☐ Cold start users

- Explicit vs implicit feedback

	Matrix	Titanic	..	Forrest Gump	Wall-E
Alice		3	..	4	
John	3		..		3
..	..	..	..	..	..
Eric	3	3	..	5	
Diane		5	..		1

	Matrix	Titanic	..	Forrest Gump	Wall-E
Alice		1	..	1	
John	1		..		1
..	..	..	..	..	..
Eric	1	1	..	1	
Diane		1	..	1	1

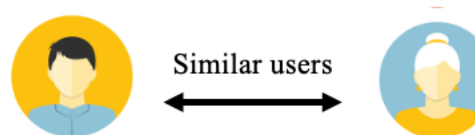
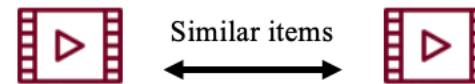
K Nearest Neighbourhood algorithms

Algorithms:

- Content Based
- User Based
- Item Based

Similarity measures

- Dice Coefficient
- (Adjusted) Cosine similarity
- Pearson's correlation coefficient



Other algorithms

- Content-based: **Naive Bayes**

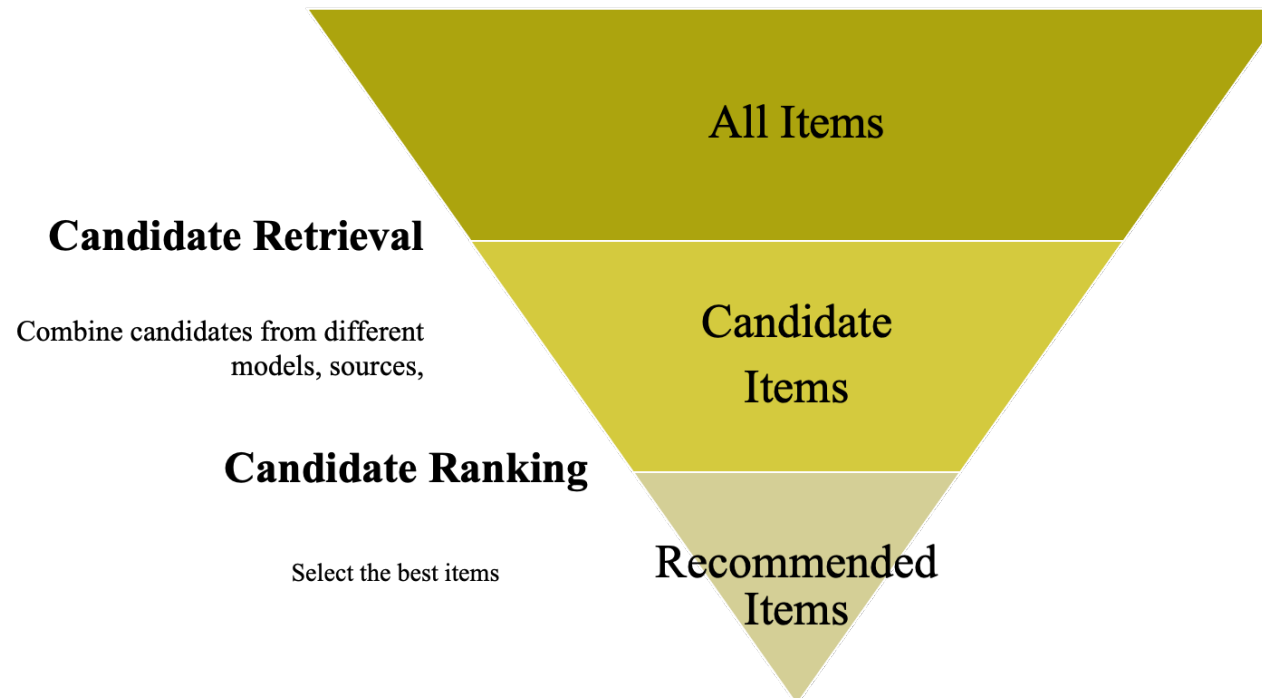
$P(\text{Like} \mid \text{Action \& SF \& Comedy \& NOT Romance \& NOT Children})$

- Collaborative Filtering: **Matrix Factorization** (using SGD)

		<i>Items</i>					
		<i>1</i>	<i>2</i>	...	<i>i</i>	...	<i>m</i>
<i>Users</i>	<i>1</i>	5	3		1	2	
	<i>2</i>		2				4
	:			5			
	<i>u</i>	3	4		2	1	
	:					4	
	<i>n</i>			3	2		
	<i>a</i>	3	5		?	1	

Other topics

- Evaluating recommenders
- Hybrid models



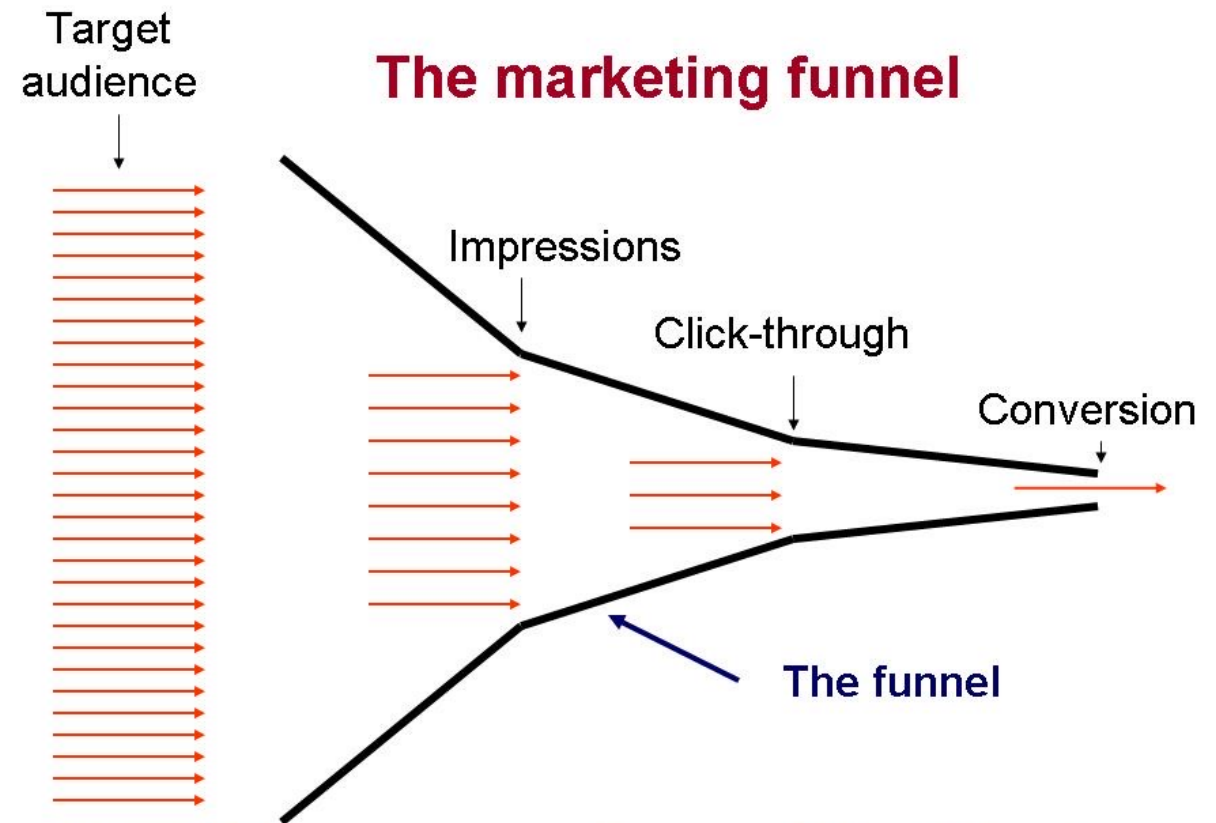


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Online Advertising

Online Advertising

- Funnel & Metrics
- Online advertising channels
 - ☐ Search Engines (SEA)
 - ☐ Social media
 - ☐ Display (RTB)
- System & Data
 - ☐ Web Analytics
 - ☐ Cookies: alternative for 3rd party cookies



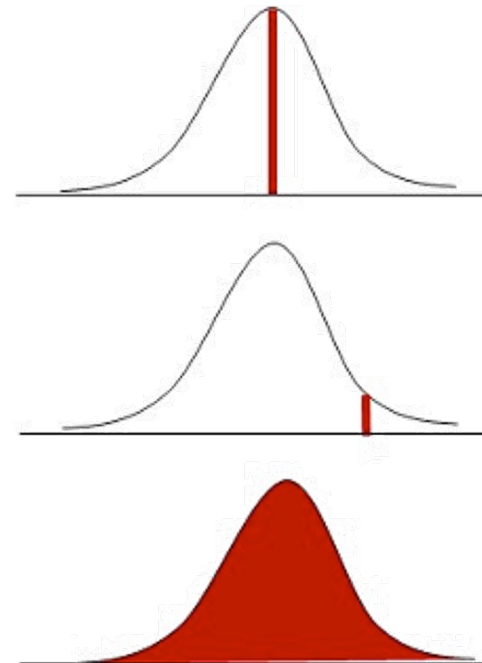
CTR Prediction

Overall click rate for a few versions:

- **Bayesian updating** to improve estimates in start-up phase
- **Reinforcement Learning** to balance exploration vs exploitation:

- **Multi-armed Bandits**

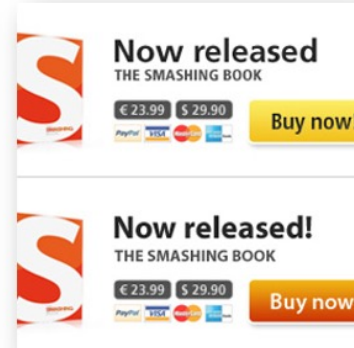
- ϵ -Greedy
- Upper Confidence Bound (UCB)
- Thompson Sampling



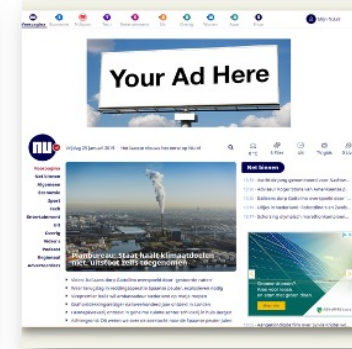
CTR Prediction



Visitor



Advertisement



Publisher



Time

Estimating click rates with a lot of features and observations (online learning)

- Naïve Bayes
- Logistic Regression using SGD
- Error Metric: Log Loss

Multichannel Advertising

■ Media Mix Modeling

- ❑ Wider coverage of (offline) channels and sales
- ❑ Higher level, longer term
- ❑ Regression model

■ Conversion Attribution

- ❑ Online channels
- ❑ Rule based vs Data Driven
- ❑ Marginal Contribution

Media Mix Modeling

Aggregate data on media spend and sales

“We advertised on TV last week and sales went up”

Conversion Attribution

Detailed data on customer touchpoints and conversions

“This person saw my advertisement on nu.nl yesterday and then entered my website. Today he came back and bought something”



Exam

- Digital (TestVision)
- Physical on a university location
- University supplies laptops
- You are allowed to use:
 - ☐ Calculator (simple, no internet connection)
 - ☐ Formula sheet (supplied by university)
 - ☐ Scratch paper (supplied by university)
- You are NOT allowed to use:
 - ☐ Connected devices: phones etc.
 - ☐ Other applications, browser tabs on the laptop